

DCCN (CS24305) 10–Page Master Quick Revision

High–Yield Formulas, Protocol Summaries, Comparison Matrices & Exam Flashcards

⚡ 10–Page Master Quick Revision — Data Communication & Computer Networks (CS24305)

Page 1: 52–Topic Master Syllabus Progress Checklist (Modules I – V)

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Page 10: TCP 3–Way Handshake, Congestion Control (AIMD) & Application Protocols

CS24305 DCCN — All 52 Topics Verified & Mastered

Module	Topic #	Topic Name	Key Focus	
M1: Overview (13)	1-3	Communication Model & Networks	Simplex/Half/Full-Duplex, Performance Criteria	
	4-5	Topologies & Categories	Mesh, Star, Bus, Ring, LAN, MAN, WAN	
	6-8	Internet, Standards & Layering	ISP Hierarchy, OSI vs TCP/IP Encapsulation	
	9-10	OSI 7-Layer & TCP/IP Suite	PDUs, Headers & 4-Level Addressing	
	11-12	Impairments & Channel Capacity	Attenuation, Nyquist Bit Rate & Shannon Capacity	
	13	Transmission Media	UTP, STP, Coax, SMF, MMF, Radio, Microwave	
	M2: Encoding (6)	14	Digital Line Coding	NRZ-L, NRZ-I, Bipolar AMI, Manchester
15		Scrambling & Block Coding	4B/5B, B8ZS (TI) & HDB3 (EI) Substitution	
16		Digital Modulation	ASK, FSK, PSK, QPSK, 16-QAM Constellations	
17		Analog to Digital (PCM/DM)	Nyquist Sampling, Quantization, SQNR	
18-19		Multiplexing (FDM/TDM)	FDM, WDM, Synchronous TDM, Statistical TDM	
M3: Data Link (8)		20-21	Framing & Transmission	Async/Sync, Byte Stuffing, Bit Stuffing HDLC
		22-23	Error Detection	Single/Burst Errors, 2D Parity, Checksum
		24	Cyclic Redundancy Check (CRC)	Modulo-2 Division, Generator Polynomials
		25-26	Hamming Error Correction	$d_{\min} \geq 2t + 1$, (7,4) Hamming Code Syndrome
		27	Flow & Error Control ARQ	Stop-and-Wait, Go-Back-N, Selective Repeat η
	M4: Switching & LANs (9)	28-29	Switching Techniques	Circuit vs Packet Switching, Datagram vs VC
		30	Switch Architectures	Crossbar, Banyan & TDM Switches
		31-32	Random Access Protocols	Pure/Slotted ALOHA, CSMA (1/non/p-persistent)
33		CSMA/CD (Ethernet)	Collision Detection, Min Frame Size, Backoff	

Module	Topic #	Topic Name	Key Focus		
34		CSMA/CA (Wi-Fi)	Hidden Terminal, RTS/CTS Handshake, NAV		
35–36		Ethernet & Layer 2 Devices	IEEE 802.3 Frame, Bridges, Switches, STP		
M5: Network, Transport, App (16)		37–38	IPv4 & IPv6 Headers		20B IPv4 Fields, Fragmentation, 40B IPv6
		39–41	Subnetting, ARP & ICMP		CIDR Prefix Math, ARP Cache, Ping/Traceroute
		42–44	Routing Protocols		Distance Vector (RIP), Link State (OSPF), BGP
	45–46	UDP & TCP Headers	8B UDP, 20B TCP Segment Flags (SYN/ACK/FIN)		
	47–49	TCP Handshake & Congestion	3–Way SYN, 4–Way FIN, AIMD, Slow Start, Reno		
	50–52	Application Layer Protocols	DNS Resolution, SMTP/IMAP, HTTP/1.1 vs HTTP/2/3		

Page 2: The Tier-S High-Yield Core Topic Ranking & Exam Strategy

Rank	Core Exam Topic	Module	Frequency & Yield	Essential Mathematical Formula / Rule
★ 1	Shannon & Nyquist Theorems	M1	100% Exam Probability (10M)	$C_{\text{Nyquist}} = 2B \log_2(L)$, $C_{\text{Shannon}} = B \log_2(1 + \text{SNR})$
★ 2	Line Coding & Manchester Waveforms	M2	100% Exam Probability (8M)	NRZ-L, NRZ-I, Bipolar AMI, Manchester (0 = H → L, 1 = L → H), Diff. Manchester
★ 3	CRC Polynomial Division	M3	100% Exam Probability (10M)	Modulo-2 Division of $D \cdot 2^r$ by $G(x)$ to get Remainder R ; Codeword $T = D \cdot 2^r + R$
★ 4	Hamming Error Correction Code	M3	100% Exam Probability (10M)	$2^r \geq m + r + 1$, $d_{\min} \geq 2t + 1$; Parity bits at powers of 2 (1, 2, 4, 8)
★ 5	Sliding Window Efficiency (η)	M3	100% Exam Probability (10M)	$\eta = \frac{W}{1+2a}$, $a = \frac{T_{\text{prop}}}{T_{\text{trans}}}$; GBN ($W = 2^k - 1$), SR ($W = 2^{k-1}$)
★ 6	ALOHA vs CSMA/CD Min Frame Size	M4	100% Exam Probability (10M)	$S_{\text{Pure}} = Ge^{-2G}$, $S_{\text{Slotted}} = Ge^{-G}$; CSMA/CD: $\text{Frame}_{\min} \geq 2T_{\text{prop}} \times \text{Bandwidth}$
★ 7	IPv4 Header & Fragmentation	M5	100% Exam Probability (10M)	$\text{Total Length} \leq \text{MTU}$, $\text{Fragment Offset} = \text{Data Offset} / 8$
★ 8	CIDR Subnetting & IP Allocations	M5	100% Exam Probability (10M)	Borrow k bits: New mask $/n + k$, Usable hosts = $2^{(32 - \text{prefix})} - 2$
★ 9	Dijkstra vs. Bellman-Ford Routing	M5	100% Exam Probability (10M)	Dijkstra Global Shortest Path; Bellman-Ford $D_x(y) = \min_v \{c(x, v) + D_v(y)\}$
★ 10	TCP 3-Way Handshake & AIMD	M5	100% Exam Probability (10M)	SYN → SYN-ACK → ACK; Slow Start ($\times 2$), Congestion Avoidance ($+1$), Fast Retransmit

The 7 OSI Layers & Protocol Units (PDUs):

1. **Physical (Bits):** Bitstream transmission, electrical voltages, pinouts.
2. **Data Link (Frames):** Hop-by-hop node delivery, MAC addressing, CRC error check.
3. **Network (Packets):** Host-to-host routing, logical IPv4/IPv6 addressing.
4. **Transport (Segments/Datagrams):** Process-to-process port delivery (TCP/UDP).
5. **Session (Data):** Dialogue control, synchronization checkpoints.
6. **Presentation (Data):** Character translation (UTF-8), compression, SSL/TLS encryption.
7. **Application (Messages):** User protocols: HTTP, DNS, SMTP, FTP, SSH.

Channel Capacity Formulas:

- Nyquist Formula (Noiseless): $C = 2B \log_2(\mathbf{L})$ bps
- Shannon Formula (Noisy): $C = B \log_2(\mathbf{1} + \text{SNR})$ bps
- Signal-to-Noise Ratio (dB): $\text{SNR}_{\text{dB}} = 10 \log_{10}(\text{SNR}) \implies \text{SNR} = 10^{\text{SNR}_{\text{dB}}/10}$
- Attenuation (dB): $\text{dB} = 10 \log_{10}(\mathbf{P}_2/\mathbf{P}_1) = 20 \log_{10}(\mathbf{V}_2/\mathbf{V}_1)$

Line Coding	Bit 0 Representation	Bit 1 Representation	Clock Sync?
NRZ-L	Positive Voltage ($+V$)	Negative Voltage ($-V$)	No
NRZ-I	No transition at start	Transition at start	On 1s only
Bipolar AMI	Zero Volts ($0V$)	Alternating $+V$ and $-V$	On 1s only
Manchester	High-to-Low at midpoint	Low-to-High at midpoint	100% Perfect
Diff. Manchester	Transition at start	No transition at start	100% Perfect

CRC Division & Error Correction Rules:

- **CRC Remainder:** Remainder of $(D \cdot 2^r)/G(x)$ using Modulo-2 XOR division.
- **Hamming Redundancy:** $2^r \geq m + r + 1$ (m data bits, r parity bits).
- **Detection Limit:** $d_{\min} \geq d + 1$; **Correction Limit:** $d_{\min} \geq 2t + 1$.

Protocol	Sender Window (W_S)	Receiver Window (W_R)	Efficiency (η)
Stop-and-Wait ARQ	1	1	$\eta = \frac{1}{1 + 2a}$
Go-Back-N (GBN)	$2^k - 1$	1	$\eta = \frac{W_S}{1 + 2a}$
Selective Repeat (SR)	2^{k-1}	2^{k-1}	$\eta = \frac{W_S}{1 + 2a}$

MAC Sublayer Formulas:

- **Pure ALOHA:** $S = Ge^{-2G} \implies S_{\max} = 18.4\%$ (at $G = 0.5$).
- **Slotted ALOHA:** $S = Ge^{-G} \implies S_{\max} = 36.8\%$ (at $G = 1.0$).
- **CSMA/CD Min Frame Size:** $\text{Frame Size}_{\min} \geq 2 \times T_{\text{prop}} \times \text{Bandwidth}$.

Subnetting & Routing Equations:

- **Usable Hosts:** $2^{(32-\text{Prefix})} - 2$
- **Fragment Offset:** $\text{Offset} = \text{Byte Index} / 8$
- **Bellman-Ford Equation:** $D_x(y) = \min_v \{c(x, v) + D_v(y)\}$
- **Dijkstra Shortest Path:** Greedy node selection with minimum cumulative edge cost.

TCP Flow & Congestion Control:

- **Slow Start:** Exponential growth: $\text{cwnd} \leftarrow \text{cwnd} \times 2$ per RTT.
- **Congestion Avoidance:** Linear growth: $\text{cwnd} \leftarrow \text{cwnd} + 1$ MSS per RTT.
- **Fast Retransmit & Recovery:** 3 Duplicate ACKs $\implies \text{ssthresh} = \text{cwnd}/2$, $\text{cwnd} = \text{ssthresh} + 3$.